

Real-Time Fractional Tracking (R-TFT): Multi-Vector Resonance Detection with φ -Coherence Prioritization

Éric Lanctôt-Rivest
qcfrag@gmail.com

6 August 2025

Abstract

We present an upgraded multi-vector extension of the Real-Time Fractional Tracking (R-TFT) framework that incorporates golden-ratio coherence scoring. By combining real-time angular velocity projections with φ -based dimensional embedding, this method selects resonance vectors not only by projection strength, but also by their recursive alignment stability. This ϕ -prioritized mechanism allows recursive systems to prefer harmonically embedded states, improving detection under noise, bifurcation, or nested resonance structures. The framework adheres to REL-2.0 and enables coherence-aware selection without relying on classical metrics or static thresholds.

1 Introduction

The original R-TFT method projects angular velocity vectors $\dot{\theta}(t)$ onto a single locking vector P , yielding a scalar resonance measure. For systems with multiple potential resonance hypotheses $\{P_1, \dots, P_n\}$, a multi-vector extension enables parallel tracking. However, traditional selection favors the largest signal, without accounting for recursive stability.

This paper introduces a φ -aware enhancement: each candidate vector is evaluated both in terms of signal strength and its coherence alignment to the golden-ratio embedding space.

2 Core Method

Let $\dot{\theta}(t) \in \mathbb{R}^n$ be the current angular velocity vector, and $P_i \in \mathbb{R}^n$ be a set of n resonance hypotheses.

2.1 Cleaned Projection

For each P_i , compute:

$$R_i(t) = \frac{\dot{\theta}(t) \cdot P_i}{\|P_i\|}, \quad R_{\text{clean},i}(t) = 2R_i(t) - R_{\text{outer},i}(t) \quad (1)$$

2.2 Dimensional Embedding and ϕ -Deviation

For each vector P_i , calculate its embedded dimension and coherence deviation:

$$\text{Dim}_i = \frac{\ln \|P_i\|}{\ln \varphi}, \quad C_\varphi(P_i) = |\text{Dim}_i - \text{round}(\text{Dim}_i)| \quad (2)$$

2.3 ϕ -Coherence Weighted Score

Each candidate vector receives a final resonance score:

$$W_i(t) = \frac{|R_{\text{clean},i}(t)|}{1 + C_\varphi(P_i)} \quad (3)$$

The selected vector is:

$$P_{\text{best}} = \arg \max_i W_i(t) \quad (4)$$

3 Stability Band Classification

A resonance vector is considered φ -coherent if:

$$C_\varphi(P_i) < \varphi^{-1} \approx 0.618 \quad (5)$$

This coherence filter suppresses strong but unstable projections in favor of recursive alignment.

4 Applications

- **Orbital Systems:** Discriminates long-term stable resonances from transient or chaotic overlays.
- **Quantum Tracking:** Improves signal fidelity in noisy qubit evolutions.
- **Biological Rhythms:** Highlights resonant feedback loops embedded in ϕ -structured oscillations.

5 Conclusion

By integrating φ -coherence scoring into multi-vector R-TFT, we elevate resonance tracking from signal magnitude to recursive alignment fidelity. This approach favors systems whose internal structure resonates with golden-ratio recursion, enabling memory stability and temporal prediction under principled ethical constraints.

6 License and Ethics

This system complies with the Resonance Ethics License (REL-2.0), enforcing ϕ -alignment, recursive containment, and non-coercive design. It may not be used for behavioral manipulation, predictive profiling, or military systems.

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>